INTERNCIRCLE CYBERSECURITY TRACK • PROJECT 04

Home Network Security Checklist & Audit Report

Analyst: THRIGUNA G.A | **Scope:** Owned Residential Wi-Fi & Endpoint Devices | **Standard:** CISA Home Network Security Guidelines

1. Authorization & Baseline Rollback Record

Rules of Engagement Compliance: In accordance with InternCircle internship rules of engagement, this assessment was conducted strictly on my own family residential wireless router and personal household devices with explicit household permission. Prior to modifying any setting, a full configuration backup file (`router_baseline_backup.bin`) was exported and original parameters were recorded below to guarantee safe, reversible recovery.

Baseline Parameter	Original Value Recorded Before Changes	Rollback Procedure
Router Management IP	`192.168.1.1` (Local LAN Gateway)	Restore via saved config file
Primary Wi-Fi SSID	`[REDACTED_ISP_DEFAULT_2.4G]` / `[REDACTED_5G]`	Re-apply previous SSID strings
Security & Encryption	Mixed WPA/WPA2-Personal (TKIP+AES)	Revert wireless security mode
Admin Credentials & Access	Factory default admin password; Remote WAN Web UI enabled	Hardware reset button (last resort)
Firmware Build	`v2.1.4_Build20230914` (Outdated by 2 major releases)	Firmware updates kept permanently

2. Connected Device Inventory & Triage

The router's DHCP client lease table was audited to catalog every active and recently connected endpoint. Each MAC address was cross-referenced against physical household hardware labels; sensitive MAC octets and local hostnames have been redacted for privacy.

#	Assigned Device Label	Masked MAC / IP	Network Segment	Verification & Risk Status
01	Analyst Primary Laptop	`XX:XX:XX:4A:19` (.102)	Primary 5 GHz WLAN	**Verified Owned** — Auto-updated
02	Analyst Smartphone	`XX:XX:XX:8C:03` (.104)	Primary 5 GHz WLAN	**Verified Owned** — Auto-updated
03	Family Member Smartphone A	`XX:XX:XX:2F:71` (.105)	Primary 5 GHz WLAN	**Verified Owned** — Auto-updated

#	Assigned Device Label	Masked MAC / IP	Network Segment	Verification & Risk Status
04	Family Member Smartphone B	`XX:XX:XX:9D:44` (.108)	Primary 2.4 GHz WLAN	**Verified Owned** — Auto-updated
05	Home Office Desktop PC	`XX:XX:XX:6B:50` (.110)	Wired Ethernet (LAN 1)	**Verified Owned** — Auto-updated
06	Living Room Smart TV (2019)	`XX:XX:XX:1E:88` (.114)	Moved to Guest Wi-Fi	**End-of-Life OS** — Isolated
07	`Unknown-Client-7E`	`XX:XX:XX:7E:12` (.119)	Identified: Wi-Fi Printer	**Resolved & Labeled**

3. Router Hardening: Before & After Checklist

Guided by CISA home network hardening recommendations, eight core router and endpoint controls were audited and updated. All passphrases and sensitive network identifiers are masked in the evidence column.

Security Item	Before (Baseline)	After (Hardened State)	Evidence & Verification
Wi-Fi Network Name (SSID)	Default ISP model string revealing router vendor	Custom non-identifying household SSID (`[REDACTED_HOME]`)	Wireless Basic Settings capture; SSID broadcast active
Wi-Fi Passphrase	Default 8-character alphanumeric sticker password	22-character 4-word random diceware passphrase	Wireless Security capture (passphrase masked)
Wireless Encryption	Mixed WPA/WPA2-Personal (allows legacy TKIP)	**WPA2-PSK [AES] / WPA3-SAE** Transitional Mode	Router Wireless Security status confirms AES/SAE only
Router Admin Password	Factory default admin credentials printed on label	Unique 18-character password stored in password manager	Admin login test confirmed old password rejected
Remote Administration	Enabled on WAN port (exposed to public internet)	**Disabled** (Local LAN management access only)	Administration > Remote Management capture
WPS (Wi-Fi Protected Setup)	Enabled (vulnerable to offline PIN brute-force)	**Disabled** (Push-button and PIN methods off)	Wireless WPS settings capture showing Disabled
Router Firmware	Outdated (`v2.1.4_Build20230914`)	Updated to latest vendor release (`v2.4.1_Build20260820`)	System Tools > Firmware Upgrade status capture

Security Item	Before (Baseline)	After (Hardened State)	Evidence & Verification
Guest Wi-Fi Network	Disabled / Not configured	Enabled ([REDACTED_GUEST]) with AP Isolation ON	Guest Network panel; Smart TV & visitors migrated

4. Sanitized Router Console Evidence Captures

EVIDENCE CAPTURE A — WIRELESS & GUEST NETWORK CONFIGURATION (POST-CHANGE)	GATEWAY: 192.168.1.1
Primary SSID (2.4 GHz / 5 GHz):	[redacted] (Vendor name removed)
Security Mode & Cipher:	**WPA2-Personal (AES) / WPA3-Personal** (TKIP Disabled)
Pre-Shared Passphrase:	[redacted] (22 chars, 4-word passphrase)
WPS Status:	**Disabled**
Guest Network (2.4 GHz):	**Enabled** — SSID: [redacted] | Client Isolation: **Enabled** | Intranet Access: **Blocked**

EVIDENCE CAPTURE B — SYSTEM ADMINISTRATION & FIRMWARE STATUS (POST-CHANGE)	SESSION: LOCAL LAN HTTPS
Administrator Account Password:	Modified from factory default (Stored offline in encrypted vault)
Remote Web Management (WAN):	**Disabled** (Port 80/443 closed on external WAN interface)
UPnP (Universal Plug and Play):	**Disabled** (Prevents silent automatic port forwarding)
Current Firmware Version:	v2.4.1_Build20260820 — Status: **Up to Date (Verified)**

5. Endpoint Updates & Backup Verification

Securing the perimeter router must be paired with host-level hygiene across all primary household computers and mobile phones:

Device Category	Automatic OS & App Updates	Data Backup Verification
Laptop & Desktop PC	Automatic OS security patches, browser updates, and Defender definitions enabled and verified current.	3-2-1 backup verified: weekly external USB drive image + encrypted cloud sync for documents.
Smartphones (3 Devices)	Automatic system security updates and app store auto-updates confirmed active on Wi-Fi.	Encrypted cloud device backups enabled; last successful backup verified within 24 hours.

Device Category	Automatic OS & App Updates	Data Backup Verification
Smart TV & Wi-Fi Printer	Printer firmware updated to latest build; Smart TV OS confirmed unsupported by manufacturer.	No personal files stored locally; Smart TV isolated on Guest Wi-Fi without LAN access.

6. Biggest Risk Identified & Remaining Actions

> **Biggest Risk Found During Audit:**
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> The router had **Remote Web Management enabled alongside the factory-default administrator password**, while running firmware over two years out of date. Any automated internet scanner probing the home's public IP address could have logged into the router control panel, altered DNS settings to redirect banking traffic, or exposed internal devices. Disabling WAN remote administration, rotating the admin credential, and patching the firmware immediately eliminated this external takeover vector.

Honest Remaining Risks & Planned Actions

- **End-of-Life Smart TV Operating System (Medium Residual Risk):** The 2019 living room Smart TV no longer receives vendor security patches. *Mitigation & Plan:* It has been moved to the isolated Guest Wi-Fi network (blocking access to laptops and phones on the main LAN). Within 6 months, we plan to disable the TV's built-in Wi-Fi and connect an actively supported external streaming stick instead.
- **WPA2/WPA3 Mixed Compatibility Mode (Low Residual Risk):** Because the older wireless printer does not support pure WPA3-SAE, the primary network runs in WPA2/WPA3 transitional mode rather than WPA3-only. *Mitigation & Plan:* A strong 22-character passphrase prevents offline dictionary attacks; once the printer is upgraded, we will switch the primary SSID to strict WPA3-Only mode.
- **Scheduled Quarterly Audit & Backup Drill (Ongoing Action):** Set a recurring 90-day calendar reminder to inspect the DHCP device list for unknown MAC addresses, verify router firmware updates, and test restoring a sample file from the offline backup drive.